

Phase 1: Methodological Limitations, Biological Criticisms & Future Iterations

An analytical audit of **Phase 1** in the Question 5 Patient Stratification pipeline, detailing methodological constraints, biological criticisms, and actionable technical solutions for future iterations.

1. Executive Summary & Audit Rationale

While Phase 1 successfully compresses ~19,757 raw transcriptomic features into an interpretable 26-feature multi-modal panel, several implicit assumptions constrain diagnostic sensitivity and spatial accuracy.

2. In-Depth Analysis of Limitations & Criticisms

1. Methodological Limitation: Marker Averaging vs. Constrained Linear Unmixing

- Current Approach:** Transcriptomic cell deconvolution (`compute_cell_deconvolution`) computes relative infiltration scores by taking the sample-wise mean log-expression across 3–5 marker genes per cell type.
- Scientific Criticism:** It does not solve a formal constrained linear unmixing or support vector regression problem (such as CIBERSORTx or EPIC: $Y = X \cdot B$, subject to $\sum B = 1, B \geq 0$). Marker averaging produces **unbounded, arbitrary score units** rather than true, non-negative cell-type percentage proportions (0–100%).
- Biological Impact:** Marker genes such as `TNF` or `TGFB1` are expressed across multiple cell types (e.g. tumour cells, endothelial cells, or activated lymphocytes), inducing cross-cell-type expression spillover and potential correlation artifacts.

2. Biological Limitation: Spatial Blindness of Bulk RNA-Seq

- Current Approach:** Bulk RNA-seq homogenises the entire tumour biopsy core into a single average expression profile prior to sequencing.
- Scientific Criticism:** Bulk transcriptomics cannot distinguish between an **Immune-Excluded** tumour (where CD8+ T cells and CAFs are both abundant, but T cells are physically trapped in the stroma surrounding the tumour margin) vs. an **Immune-Infiltrated (Hot)** tumour (where T cells physically infiltrate the malignant cell nest).
- Clinical Impact:** Two patients with identical bulk `CD8_T_cells` and `CAFs` scores may experience completely opposite clinical outcomes due to spatial microenvironmental architecture that bulk sequencing inherently erases.

3. Technical Limitation: Absence of Direct Copy Number Alteration (CNA) Maps

- Current Approach:** Trial cohorts ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#)) lack processed GISTIC arm-level Copy Number Alteration (CNA) files.
- Scientific Criticism:** Chromosomal instability (high aneuploidy burden) is a major driver of immune exclusion and anti-PD-1 (`CD274`) resistance. Phase 1 relies on Whole Exome Sequencing (WES) mutational burden (`TMB_NONSYNONYMOUS`) and single-gene transcript levels (`PTEN` , `CDKN2A`) as indirect proxies.
- Data Impact:** Tumour Mutational Burden (TMB) and CNA burden can be decoupled (e.g. high-CNA, low-TMB tumours exist), leaving a potential blind spot for chromosomal structural variation.

4. Biological Oversimplification: 1D Macrophage Polarisation Axis

- Current Approach:** The Macrophage STV score collapses macrophage biology into a 1D scalar balance (`M1_M2_Ratio`).
- Scientific Criticism:** In human tumours, Tumour-Associated Macrophages (TAMs) exist along a continuous, multi-dimensional functional spectrum (e.g. M2a, M2b, M2c, tissue-resident, lipid-laden) rather than a strict binary M1 (antitumour) vs. M2 (pro-tumour) axis.

5. Sampling Limitation: Static Pre-Treatment Snapshots

- Current Approach:** Phase 1 feature matrices are constructed exclusively from baseline, pre-treatment biopsies.
- Scientific Criticism:** Tumour microenvironments are dynamic and evolve rapidly under therapeutic pressure (e.g. on-treatment T-cell recruitment, acquired resistance mutations, and adaptive `CD274` / PD-L1 upregulation). A static pre-treatment snapshot cannot capture early on-treatment immune dynamics occurring 2–4 weeks post-infusion.

Summary of Impact on Downstream Phases

Unbounded deconvolution scores and spatial blindness can propagate unmeasured variance into Phase 3 unsupervised clustering and Phase 5 subgroup predictive models, leading to potential misclassification of stroma-excluded tumours as immune-hot.

3. Proposed Fixes & Strategic Roadmap for Future Project Iterations

To address these limitations in future iterations of Question 5, we propose five concrete technical enhancements:

Proposed Fix 1: Transition to CIBERSORTx / Single-Cell Reference Deconvolution

- Technical Solution:** Replace marker gene averaging with CIBERSORTx or EPIC support vector regression using single-cell RNA-seq (scRNA-seq) melanoma reference matrices (e.g. Jerby-Arnon et al. or Tirosh et al.).
- Expected Benefit:** Yields true, absolute cell-type fractions (0–100%) subject to sum-to-one constraints, eliminating marker spillover and arbitrary scaling units.

Proposed Fix 2: Integration of Spatial Transcriptomics & Multiplex Immunofluorescence (mIF)

- **Technical Solution:** Incorporate spatial transcriptomics (10x Visium / Xenium) or 7-colour multiplex immunofluorescence (mIF) image quantification.
- **Expected Benefit:** Formally calculates a **Spatial Infiltration Distance Metric** (distance from CD8+ T cells to malignant cells vs distance to CAFs), enabling the model to explicitly separate *Immune-Excluded* from *Immune-Infiltrated* phenotypes.

Proposed Fix 3: Direct WGS / SNP-Array Processing for Arm-Level CNA & Aneuploidy

- **Technical Solution:** Process raw BAM/CRAM alignment files using CNVkit or ASCAT to generate explicit arm-level Copy Number Alteration (CNA) fractional loss/gain vectors and whole-genome duplication (WGD) indicators.
- **Expected Benefit:** Directly quantifies chromosomal instability independent of TMB, closing the proxy gap for immune exclusion driven by aneuploidy.

Proposed Fix 4: Multi-State Macrophage Deconvolution via Single-Cell Signatures

- **Technical Solution:** Expand the 1D STV score into a 4-dimensional macrophage vector capturing distinct subtypes: TAM_{Pro-inflammatory} (M1), TAM_{Angiogenic} (M2a), TAM_{Immunosuppressive} (M2c), and TAM_{Lipid-Laden}.
- **Expected Benefit:** Captures functional macrophage plasticity and resolves fine-grained immunosuppressive mechanisms.

Proposed Fix 5: Longitudinal Paired Baseline & On-Treatment Biopsy Modelling

- **Technical Solution:** Integrate paired baseline (Day 0) and early on-treatment (Day 14–28) biopsy transcriptomics.
- **Expected Benefit:** Calculates dynamic **Delta Signatures** ($\Delta TIS = TIS_{Day28} - TIS_{Day0}$), capturing early T-cell expansion and adaptive immune reactivation that static baseline biopsies cannot observe.

4. Comprehensive Roadmap Comparison Matrix

Limitation Category	Current Phase 1 Method	Proposed Technical Fix	Primary Tool / Platform	Expected Clinical & Analytical Benefit
Cell Deconvolution	Mean marker log-expression averaging	Constrained SVR linear unmixing	CIBERSORTx / EPIC	Bounded 0–100% absolute cell fractions; no marker spillover.
Spatial Architecture	Bulk RNA-seq homogenisation	Spatial transcriptomics & mIF	10x Visium / Xenium / mIF	Accurately separates stroma-excluded from tumour-infiltrated T cells.
Genomic Instability	Indirect TMB & single-gene expression proxies	Direct WGS / SNP-array copy number callouts	CNVkit / ASCAT / GISTIC 2.0	Direct measurement of arm-level CNA and aneuploidy-driven exclusion.
Macrophage Subtypes	1D M1/M2 binary ratio (M1_M2_Ratio)	4-state single-cell macrophage vector	scRNA-seq reference matrices	Resolves M2a/M2c subtype heterogeneity and therapeutic drug targets.
Biopsy Timing	Static pre-treatment snapshot (Day 0)	Longitudinal paired biopsy modeling (Δ scores)	Paired Day 0 & Day 14–28 RNA-seq	Observes early on-treatment T-cell expansion and adaptive resistance.

 **Key Takeaways & Strategic Future Roadmap**

- **Methodological Evolution:** Moving from marker averaging to CIBERSORTx SVR unmixing will eliminate arbitrary score scaling and cross-cell spillover.
- **Spatial Resolution:** Integrating spatial transcriptomics (10x Visium) resolves the spatial blindness of bulk RNA-seq, allowing precise detection of stroma-excluded tumours.
- **Multi-Omics Precision:** Processing raw WGS/SNP-arrays for direct arm-level CNA calls will eliminate reliance on indirect TMB proxies for chromosomal instability.